SECSR2043 OPERATING SYSTEMS
[20 Marks]

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Instruction: Please answer all of the following questions. Whenever the □ symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname*.sh (Example: ahmad.sh).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd
../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar .
```

a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with the **$HOME** directory.

[4marks]

Command Line Script

```
adam@secr2043:~$ nano adam.sh
adam@secr2043:~$ chmod +x adam.sh
adam@secr2043:~$ ./adam.sh
```

```
adam@secr2043:~$ ls -R helloworld.jar cars dates fruits drinks
helloworld.jar

cars:
accord.c   civic.c   proton.c

dates:
appointment   dateoftheday

drinks:
accord.c   apple.txt   civic.c   helloworld.jar   orange.txt   proton.c

fruits:
apple.txt   orange.txt
```

Tree Graph Structure

```
$HOME
├──helloworld.jar
│   ├──cars
│   │   ├──accord.c
│   │   └──civic.c
│   └──dateoftheday
├──dates
│   ├──appointment
│   └──dateoftheday
├──fruits
│   ├──apple.txt
│   └──orange.txt
├──drinks
│   ├──accord.c
│   ├──apple.txt
│   ├──civic.c
│   ├──helloworld.jar
│   └──orange.txt
└──────proton.c
```

b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter "c" as the input to the bash script. [4 marks]

```
#!/bin/bash

# This script finds and counts files with a user-specified extension.

# 1. Prompt the user and read the input
echo "Enter a file extension (e.g., c, txt, sh):"
read extension

echo ""
echo "--- Searching for files with the .${extension} extension ---"

# 2. Find and display the files
# Using 'find' is safer than 'ls' if no files are found.
# -maxdepth 1 searches only the current directory.
# -type f ensures we only find files, not directories.
find . -maxdepth 1 -type f -name "*.${extension}"

# 3. Count the number of files found and display the count
# We store the result of the command in a variable.
count=$(find . -maxdepth 1 -type f -name "*.${extension}" | wc -l)

echo "-------------------------------------------------------"
echo "Found ${count} file(s) with the .${extension} extension."
```

```
adam@secr2043:~$ cd cars
adam@secr2043:~/cars$ chmod +x exp.sh
adam@secr2043:~/cars$ ./exp.sh
Enter a file extension (e.g., c, txt, sh):
c

--- Searching for files with the .c extension ---
./civic.c
./accord.c
./proton.c
-------------------------------------------------------
Found 3 file(s) with the .c extension.
```

2. The following Figure 1 illustrates a tree structure of some directories and files.

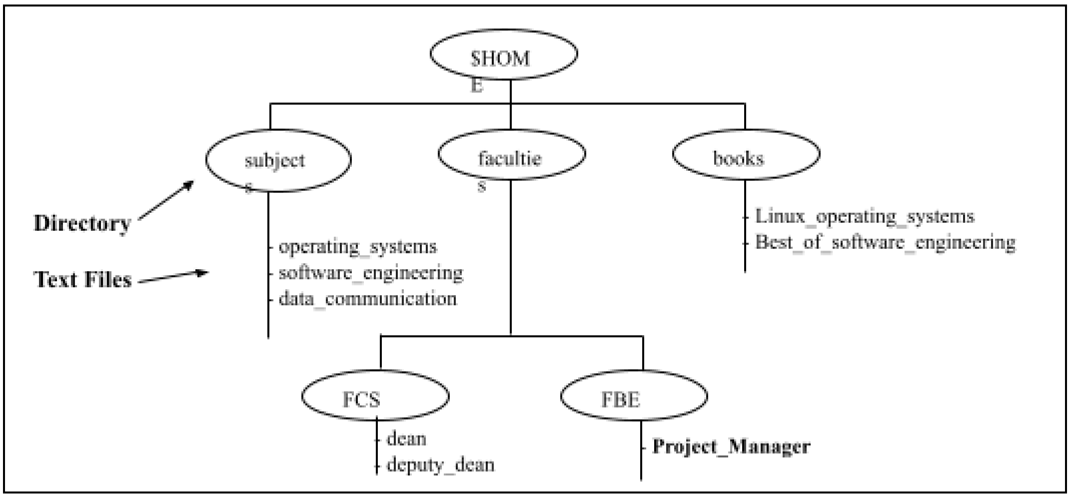


Figure 1

a) Write a bash script (called myname2a.sh) that will produce directories and files as in Figure 1. Each text file contains its filename without the underscore character**.** For example: text file Project_Manager contains Project Manager). [4 marks]

```bash
#!/bin/bash

# This script creates the directory and file structure from Figure 1.

echo "Creating directory structure..."

# Create all directories. -p flag creates parent directories if needed.
mkdir -p subject books faculti/FCS faculti/FRE

# Create files in 'subject' directory
echo "operating systems" > subject/operating_systems
echo "software engineering" > subject/software_engineering
echo "data communication" > subject/data_communication

# Create files in 'books' directory
echo "Linux operating systems" > books/Linux_operating_systems
echo "Best of software engineering" > books/Best_of_software_engineering

# Create files in 'faculti/FCS' directory
echo "dean" > faculti/FCS/dean
echo "deputy dean" > faculti/FCS/deputy_dean

# Create file in 'faculti/FRE' directory
echo "Project Manager" > faculti/FRE/Project_Manager

echo "Structure created successfully. Use 'ls -R' to view it."
```

```
adam@secr2043:~$ ls -R subject books faculti
books:
Best_of_software_engineering  Linux_operating_systems

faculti:
FCS  FRE

faculti/FCS:
ls: cannot access 'faculti/FCS/deputy_dean': Permission denied
ls: cannot access 'faculti/FCS/dean': Permission denied
dean  deputy_dean

faculti/FRE:
Project_Manager

subject:
data_communication  operating_systems  software_engineering
adam@secr2043:~$
```

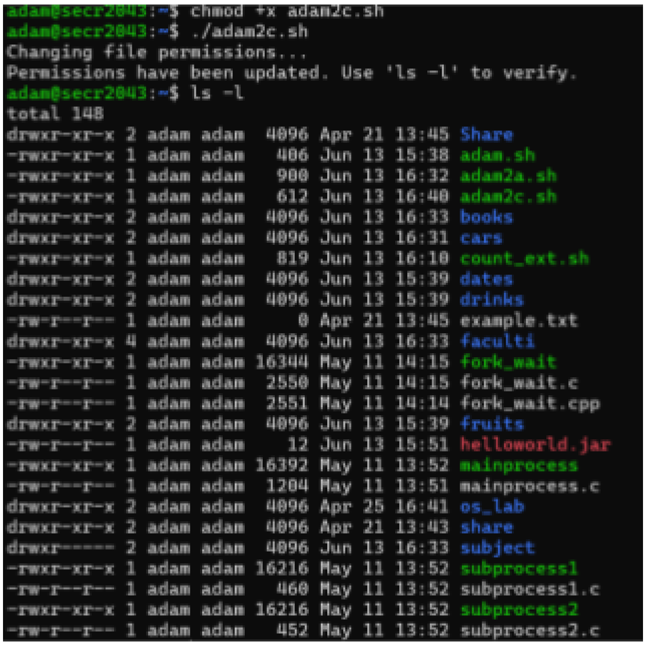
b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for the directory called book. [2 marks]

Directory/File	Access Control
books	drwxrwxr-x
subjects	drwxrwxr-x
Best_of_software_engineering	-rw-rw-r--
FCS	drwxrwxr-x
project_manager	-rw-rw-r--

c) Write another bash script (called myname2c.sh) that will change the access control of the directories and files based on the following information:

Directory/File	Users								
	Owner			Group			Public		
subjects	🚖	🚖	🚖	🚖	x	x	🚖	x	x
Best_of_software_engineering	🚖	x	🚖	x	🚖	x	x	x	x
FCS	🚖	🚖	x	x	x	x	🚖	🚖	🚖
project_manager	x	x	x	x	🚖	🚖	x	x	🚖

```
#!/bin/bash

# This script changes permissions based on the table in question 2c.
# It assumes the file structure from 2a has already been created.

echo "Changing file permissions..."

# subjects: Owner=rwx(7), Group=r--(4), Public=---(0)
chmod 740 subject

# Best_of_software_engineering: Owner=r-x(5), Group=r-x(5), Public=---(0)
chmod 550 books/Best_of_software_engineering

# FCS: Owner=rw-(6), Group=---(0), Public=rwx(7)
chmod 607 faculti/FCS

# Project_Manager: Owner=---(0), Group=rw-(6), Public=--x(1)
chmod 061 faculti/FRE/Project_Manager

echo "Permissions have been updated. Use 'ls -l' to verify."
```



```
adam@secr2043:~$ chmod +x adam2c.sh
adam@secr2043:~$ ./adam2c.sh
Changing file permissions...
Permissions have been updated. Use 'ls -l' to verify.
adam@secr2043:~$ ls -l
total 148
drwxr-xr-x 2 adam adam  4096 Apr 21 13:45 Share
-rwxr-xr-x 1 adam adam   406 Jun 13 15:38 adam.sh
-rwxr-xr-x 1 adam adam   900 Jun 13 16:32 adam2a.sh
-rwxr-xr-x 1 adam adam   612 Jun 13 16:40 adam2c.sh
drwxr-xr-x 2 adam adam  4096 Jun 13 16:33 books
drwxr-xr-x 2 adam adam  4096 Jun 13 16:31 cars
-rwxr-xr-x 1 adam adam   819 Jun 13 16:10 count_ext.sh
drwxr-xr-x 2 adam adam  4096 Jun 13 15:39 dates
drwxr-xr-x 2 adam adam  4096 Jun 13 15:39 drinks
-rw-r--r-- 1 adam adam     0 Apr 21 13:45 example.txt
drwxr-xr-x 4 adam adam  4096 Jun 13 16:33 faculti
-rwxr-xr-x 1 adam adam 16344 May 11 14:15 fork_wait
-rw-r--r-- 1 adam adam  2550 May 11 14:15 fork_wait.c
-rw-r--r-- 1 adam adam  2551 May 11 14:14 fork_wait.cpp
drwxr-xr-x 2 adam adam  4096 Jun 13 15:39 fruits
-rw-r--r-- 1 adam adam    12 Jun 13 15:51 helloworld.jar
-rwxr-xr-x 1 adam adam 16392 May 11 13:52 mainprocess
-rw-r--r-- 1 adam adam  1204 May 11 13:51 mainprocess.c
drwxr-xr-x 2 adam adam  4096 Apr 25 16:41 os_lab
drwxr-xr-x 2 adam adam  4096 Apr 21 13:43 share
drwxr----- 2 adam adam  4096 Jun 13 16:33 subject
-rwxr-xr-x 1 adam adam 16216 May 11 13:52 subprocess1
-rw-r--r-- 1 adam adam   460 May 11 13:52 subprocess1.c
-rwxr-xr-x 1 adam adam 16216 May 11 13:52 subprocess2
-rw-r--r-- 1 adam adam   452 May 11 13:52 subprocess2.c
```

d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	drwxr-----
Best_of_software_engineering	-r-xr-x---
FCS	drw----rwx
project_manager	----rw--x

End of Lab 3

**** All the Best for Final Exam ****